

**PETUNJUK TEKNIS DAN KISI – KISI LKS SMK XXXII
TINGKAT PROVINSI JAWA TENGAH
TAHUN 2024**

02 – 04 Mei 2024

<https://lksjateng.com>

"Integritas, Kompeten dan Prestasi"



**PEMERINTAH PROVINSI JAWA TENGAH
DINAS PENDIDIKAN DAN KEBUDAYAAN
2024**

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1. Introduction to Test Project

The following is a list of sections or information that must be included in all Test Project proposals that are submitted to LKS.

- Introduction/overview
- Short description of project and tasks
- Instruction to the Competitor

2. Introduction

The target of this competition is to create assets from the concept for AAA game

a. Game Background

In a barren and unforgiving world, **Skull Rider**, an anti-hero with a skull-faced visage and cowboy attire, wages a merciless yet just battle against crime. When he arrives in a city ruled by a criminal gang, his encounter with a desperate young woman sparks his epic journey to dismantle the dark forces. Amidst conflicts fraught with murder, betrayal, and inner struggles, Skull Rider confronts his dark past while striving to find a glimmer of hope amid the prevailing darkness, leading him on a tense journey toward justice and perhaps, ultimately, peace.

b. The Brief

Through the competition, you will be creating two Assets for the game:

- Major Asset: **Skull Rider**
 - Hats, robes and pants are made of leather with gold ornaments
 - Eyes lit up
 - scarves made of cloth
- Minor Asset: **Revolver and Pirate Sword**
 - pistols and swords are made of shiny steel with gold wrapping

c. Desain project



d. Schedule Overview

The result for this Competition will include the following:

Day 1 Sculpting, Modeling dan UV Unwrapping

Day 2 Modeling, UV Unwrapping, Texturing dan Presentation

Day	DELIVERABLE	DESCRIPTION
1	Modelling	- Sculpt the character (Skull Rider) - Must be high poly and the view presentation must be from the front, left, right, 3/4 - Rough model for props (Revolver and Pirate Sword) - The model with a maximum of 10.000 triangles
2	Modeling and UV unwrapping	- Finish Character (Skull Rider) - The model with a maximum of 50.000 triangles - Finish props (Revolver and Pirate Sword) - The model with a maximum of 10.000 triangles - Unwrap Character - Unwrap Props
2	Texturing and Presentation	- Texture Character 4096 x 4096px texture map - Ambient occlusion, normal, and opacity maps - Diffuse/Metal/Rough PBR - Texture props 1024 x 1024px texture map - Ambient occlusion, normal, and opacity maps - Diffuse/Metal/Rough PBR - Presentation - The character must have posed with a weapon companion - The character and companion must be presented from the front, left, right, and 3/4 view - The character and companion must be presented from the front, left, right, and 3/4 view with a Wireframe

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDER ON THE SERVER

3. Description of project and task

a. Day 1 - sculpting, Modeling and unwrap UV

Day 1 Brief

You will be creating **Skull Rider**, and **today you need to sculpt for Character (Skull Rider) and a rough 3D model for the props (Revolver and Pirate Sword) process**. Following this, you need to create a 3D sculpt and dummy model for the competition.

The committee will provide:

- The concept art for Character and props
- The base character model (can be create from zero)
- Presentation template

Day 1 Deliverables

By the end of today, you will need to deliver :

1. 3D Sculpt for the character such as the concept art
2. The base model for the weapon companion
3. 3D Sculpt Presentation view from the front, Side, 3/4 for the character.
4. Base model presentation view from the front, Side, 3/4 for a weapon companion.
5. can continue retopology if have time

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 1 General Guidelines

Description

- 3D Sculpt for the character filename structure must be like
"CHR_NameCharacter_Sculpt"
- The base model for a weapon companion structure must be like
"PRD_NameOfWeapon_Dummy"
- the file extension must be support for Maya, 3dmax, blender, ZBrush, or related 3D software support 3D sculpt i.e .fbx, .obj, .ztl, .zbr

b. Day 2 – Modeling, unwrap UV, Texturing and Presentation

Day 2 Brief

Example for the brief

The 3D sculpt for the character and base model weapon must be complete. It is not important in which order you complete the models as long as they are both ready for marking by the end of today. The 3D sculpture character should be retopology and must remain within the 50.000 triangle limit.

The base model of the weapon companion has to be completed today. Both of them must be UV unwrapped.

Texture the character and props, character texture sheets of 4096 x 4096px (A Royal Guard) and 1024 x 1024px (A weapon companion). Do not make the assets share 1 sheet.

Make the most out of each sheet – note that marks are awarded for the effective use of the texture sheets.

Create texture for each asset, though you may also paint detail and create maps in 3d painting software such as a substance painter, photoshop, substance designer, or 3dCoat. Think about the variety of materials that may be used, for example, metal, plastic, skin, fiber, etc.

Make sure to follow the look and feel of the reference images.

Day 2 Deliverables

1. Character (**Skull Rider**) – Maximum 50.000 triangles
2. Props (**Revolver and Pirate Sword**) – Maximum 10.000 triangles
3. Both of them must be UV unwrapping
4. texture map for the character (A Royal Guard) and Props (weapon companion)
 - Diffuse map
 - Metal map
 - Rough map
 - Ambiance Occlusion (AO) map
 - Normal map
 - Opacity map
5. 2 texture sheets
 - Character – 4096 x 4096px
 - Props – 1024 x 1024px
6. Final 3D model for the character and props with the final texture
7. Character and props must be posed for presentation
8. Character and props must be a presentation from a perspective view (3/4 front and 3/4 back)

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 2 General Guidelines

Description

1. Maya, Max, or Blender Scene Organisation – only relevant assets in the scene
2. Object pivot must be in zero grid coordinate (0,0,0)
3. The pivot must in below the object
4. Unit in Scene must be 10cm/grid
5. Topology and edge loop distribution for character and props. Consider areas that require deformation
6. Both 3D models conform to edge flow that accentuates the represented asset.
7. (major/minor) Assets can be understood in silhouette only

8. No N-gons are present on the assets.
9. No flipped normal is present on the assets.
10. Chamfered edges on any 90 degrees on the assets
11. The model utilizes 90+% of the triangle budget on the assets.
12. The model is within the triangle budget
13. Character filename structure **"CHAR_nameCharacter_MDL"**
14. Props filename structure **"PROP_nameOfWeapon_Dummy"**
15. the file extension must be support for Maya, 3dmax, blender i.e .fbx, .obj
16. Aspect Ratio of texture sheets are correct **—4096x4096(major) & 1024x1024(minor)**
17. PSD saved as PNG format and applied to the mode
18. UV Tidiness and Efficiency - character (A Royal Guard)
19. Smooth and even UV shells
20. Texel density – the size of polygons relative to the pixel resolution – Character/Pops
21. Group shells with similar colors together
22. UV coordinates exported to PS or Substance
23. UV Tidiness and Efficiency – Props (weapon companion)
24. No overlapping UVs (except for mirror/repeat)
25. UVs utilize rotation to maximize space
26. No obvious pixelation on the surface
27. No obvious stretching of textures on the surface
28. Color and Tone represents the base color of the material
29. Surface Textures - describes materials correctly
30. The texture looks seamless on the model
31. The texture is consistent with an art style
32. A specular or Metal map created
33. Normal map created
34. A roughness or smoothness map created
35. A variety of physical materials have been represented
36. Appropriate use of Opacity map created for transparency
37. Ambient Occlusion map created
38. 2 or more base colors have been used on the texture sheet
39. Height or Displacement map created
40. Smart masks used in Substance Painter
41. Evidence of variation in PBR textures (settings)
42. For character, the filename must be **"CHAR_nameCharacter_FINAL"**
43. For props, the filename must be **"PROP_nameCharacter_FINAL"**
44. Texture filename for character must be :
 - a. Diffuse = **"CHAR_nameCharacter_DIFF"**,
 - b. AO = **"CHAR_nameCharacter_AO"**,
 - c. Metal = **"CHAR_nameCharacter_MTL"**

- d. Rough = "CHAR_nameCharacter_RGH"
 - e. Opacity = "CHAR_nameCharacter_OPT"
 - f. Normal = "CHAR_nameCharacter_NRM"
45. Texture filename for props must be :
- a. Diffuse = "PROP_nameCharacter_DIFF",
 - b. AO = "PROP_nameCharacter_AO",
 - c. Metal = "PROP_nameCharacter_MTL"
 - d. Rough = "PROP_nameCharacter_RGH"
 - e. Opacity = "PROP_nameCharacter_OPT"
 - f. Normal = "PROP_nameCharacter_NRM"

4. Instructions to the Competitor

Create a folder on the server provided by the committee

The structure folder example must be like in the below:

RootFolder must be named "**your sub Provice_YourName_YourSchool**"
example

Folder name : **Surakarta_agusKuncoro_SMK7Kudus**

The folder must be "**Year + Month + Day**" The example folder name must be
"20240201"

Year = 2024

Month = 02

Day = 01

Every Final progress must upload to that folder

Internet Access

- Need internet access

Remember THE STANDARDS SPECIFICATION...!!!